

Complete Automation Map

End-to-end automation topology classifying ingestion, resolution, quality control, and dispatch workflows across autonomous agents and human specialists.

 **Automation Philosophy: Autonomous Execution with Deterministic Containment**

Gieni automates 85%+ of the end-to-end research and delivery lifecycle through scheduled scrapers, multi-agent LLM analysis, and event-driven webhooks. However, where legal title, fiduciary authority, or large capital allocations are concerned, full automation without human oversight creates catastrophic error rates. Gieni enforces a three-tier classification: Fully Automated, Semi-Automated (Human-in-the-Loop), and Pure Manual Specialist Operations.

1. Pipeline Automation Matrix

STAGE	AUTOMATION LEVEL	PRIMARY ENGINE / AGENT
1. Court Docket Scraping	[FULLY AUTOMATED]	Municipal Ingestion Worker
2. Property Matching & APN	[FULLY AUTOMATED]	Property Identity & Parcel Recon
3. Deed Retrieval & Equity	[FULLY AUTOMATED]	Ownership Intelligence Engine
4. Stakeholder & Control Parse	[SEMI-AUTOMATED]	Control Intelligence Agent
5. Authority Classification	[SEMI-AUTOMATED]	Authority Resolution Engine
6. Decision-Maker Enrichment	[FULLY AUTOMATED]	Enrichment Pipeline Worker
7. Viability Scoring (0-100)	[FULLY AUTOMATED]	Opportunity Scoring Engine
8. Quality Control Gate	[SEMI-AUTOMATED]	QC Agent & Human Reviewer
9. Client Delivery & CRM Sync	[FULLY AUTOMATED]	Reporting Agent & Webhook Dispatch
10. Disposition Feedback Loop	[SEMI-AUTOMATED]	Feedback Agent & Account Exec
11. County Playbook Expansion	[MANUAL SPECIALIST]	Gieni Expansion Taskforce

2. Fully Automated Workflows (Zero Human Touch)



Automated Court Scrapers & Docket Ingestion

- Mechanism: Scheduled headless browser workers (Playwright) trigger on a daily cron at 03:00 AM local time.
- Execution: Authenticates with municipal portals, queries probate docket registries for new filings, downloads petition PDFs, and extracts docket numbers into the Counties and Probate Opportunities databases.
- SLA / Runtime: Execution time < 15 minutes per county.



Parcel Attribution & GIS Property Matching

- Mechanism: Property Identity Agent (PIA) + Parcel Reconciliation Agent (PRA).
- Execution: Matches decedent name and last known domicile against County Tax Assessor GIS vector layers. Calculates the Parcel Attribution Score (PAS). If $PAS \geq 70$, the APN, situs address, and tax assessment are automatically committed.



Opportunity Scoring & Priority Tiering

- Mechanism: Opportunity Scoring Agent (OSA).
- Execution: Computes Composite Viability Score (0–100) using the multi-variable formula balancing Net Equity against Deal Friction Score (DFS). Automatically assigns Priority A (≥ 80), Priority B (60–79), or Priority C (40–59).



Multi-Channel Dispatch & Partner CRM Webhooks

- Mechanism: Reporting Agent (RA).
- Execution: Upon QC certification approval, RA dispatches real-time webhooks to the partner's CRM (Podio / Salesforce / HubSpot), triggers SMS flash alerts for Priority A deals, and stages weekly batch records.

3. Semi-Automated Workflows (Human-in-the-Loop)

In high-friction or high-ambiguity probate scenarios, autonomous agents perform initial heavy extraction and draft structured hypotheses, then route edge cases to human specialists:



Authority Resolution & Ambiguity Review

- **Agent Action:** Extracts petitioner, nominated executor, and Letters status from probate pleadings.
- **Human Trigger:** If docket indicates competing petitions, will contests, or dependent administration requiring court confirmation.
- **Specialist Role:** Probate analyst reviews scanned codicils and updates Authority Classification to Tier 1, 2, 3, or Unresolved.



Pre-Delivery Quality Control Certification

- **Agent Action:** Quality Control Agent (QCA) performs automated 10-point cross-reference validation.
- **Human Trigger:** Any score discrepancy, conflicting deed dates, or borderline Priority A files (scores 78–82).
- **Specialist Role:** QC Lead inspects the evidence pack and toggles QC Status to Certified or pushes to Tasks & Exceptions.



Monthly Model Recalibration & Weight Adjustments

- **Agent Action:** Feedback Agent (FA) aggregates partner disposition data (appointment rates, closed wholesale fees).
- **Specialist Role:** Lead Data Scientist reviews coefficient suggestions and recalibrates county weights.

4. Pure Manual Specialist Operations

- **County Feasibility Audits:** Initial manual reconnaissance of county clerk portals, court software types, and filing volumes before automated scraper deployment.
- **Partner Onboarding & Buy-Box Calibration:** Executive strategy sessions to define equity thresholds, asset criteria, and custom CRM field mapping.
- **Legal & Regulatory Compliance Audits:** Quarterly review of public records privacy guidelines, state solicitation statutes, and DNC registry compliance.

5. Core Automation Technology Stack



Production Infrastructure Stack

- Ingestion & Scraping: Playwright / Node.js headless workers deployed on AWS ECS with residential proxy rotation.
- Relational Storage: Managed PostgreSQL (Amazon RDS) with PostGIS extensions for spatial parcel queries.
- Orchestration & Queuing: Redis BullMQ for distributed task processing and Temporal for DAG workflow coordination.
- Operational Operating System: Notion REST API v2022-06-28 powering the Gieni OS workspace databases.
- Partner Integration: REST Webhooks, SendGrid transactional email, and Twilio SMS dispatch.